Statistics and Financial Data Analysis

UNIVERSITY OF OXFORD
Mathematical Institute

MSc Mathematical and Computational Finance

Problem Sheet 2

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Oxford Mathematics

Black–Scholes and Implied Volatility

For a European call with time to maturity $\tau = T - t > 0$,

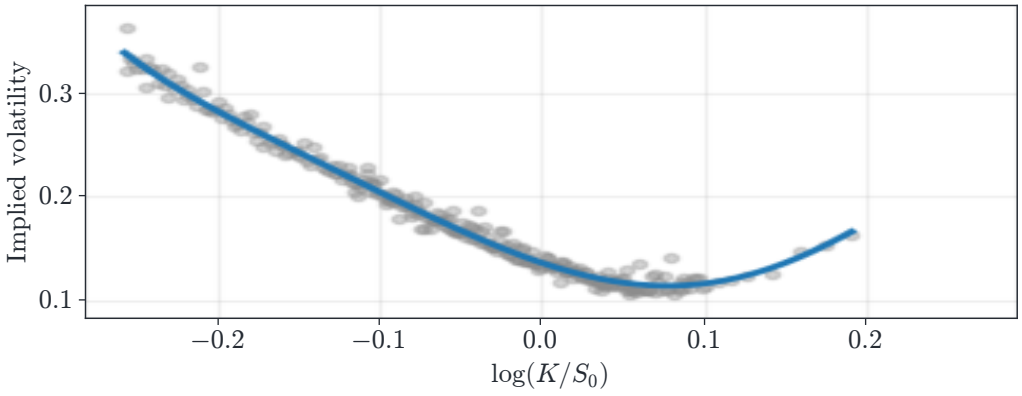
$$c_{\mathrm{BS}}(\sigma) = S_t \Phi(d_1) - K e^{-r\tau} \Phi(d_2),$$
$$d_1 = \frac{\log(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Here Φ is the standard normal distribution function.

With S_0, K, r, q, τ fixed, the price is increasing in σ.

$$c_{\mathrm{BS}}(\sigma_{\mathrm{IV}}) = c_{\mathrm{market}} \quad \Longleftrightarrow \quad \boxed{\sigma_{\mathrm{IV}} = c_{\mathrm{BS}}^{-1}(c_{\mathrm{market}})}.$$

Implied Volatility Smile

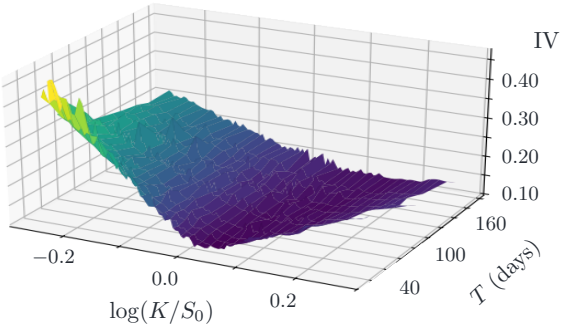


GBM predicts a flat IV smile, but market data show otherwise.

Implied Volatility Surface

The surface combines log-moneyness and maturity:

$$(x, \tau) \longmapsto \sigma_{\mathrm{IV}}(x, \tau).$$



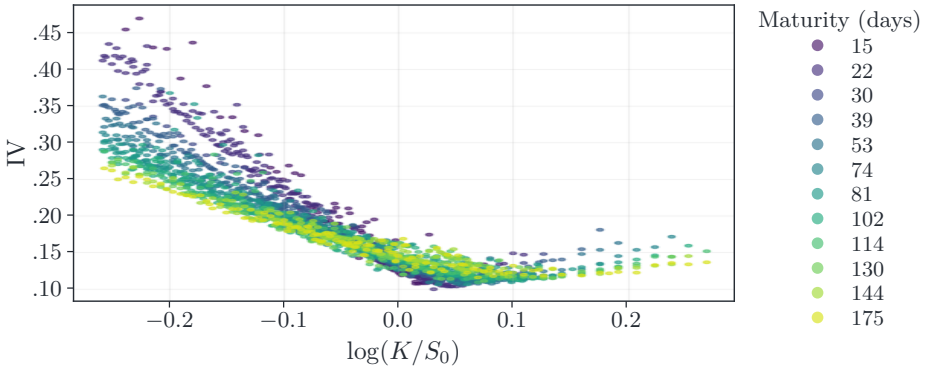
Q1(a): Data Overview

2,447 observations, 12 maturities, 15–175 days.

Variable	Mean	Std. dev.	Min.	Median	Max.
$\log(K/S_0)$	-0.050	0.100	-0.258	-0.042	0.268
Maturity (days)	76.947	43.932	15	74	175
Implied volatility	0.182	0.064	0.100	0.165	0.470

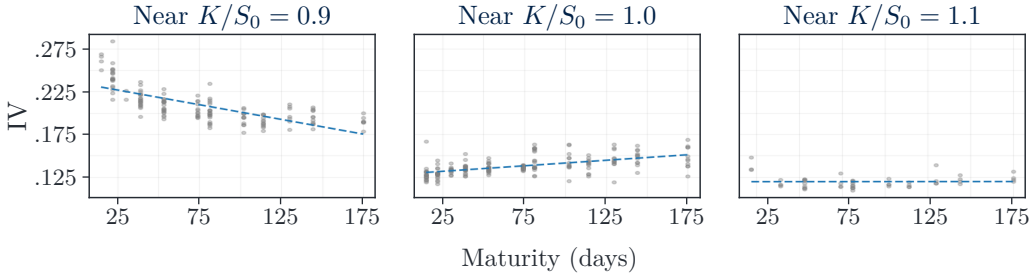
- Each maturity contains between 68 and 345 observations.
- IV is in decimal units: $0.182 = 18.2\%$.
- The available log-moneyness range differs across maturities.

Q1(a): Observed IV Smiles



IV is negatively associated with log-moneyness (pooled correlation -0.883). The slope and right-wing upturn vary with maturity.

Q1(a): The IV Term Structure



- Near $K/S_0 = 0.9$: decreasing; near 1.0: increasing; near 1.1: roughly flat.
- These are approximate slices: $|K/S_0 - m| \leq 0.01$.

The maturity pattern depends on moneyness.

Q1(b): Polynomial Regression

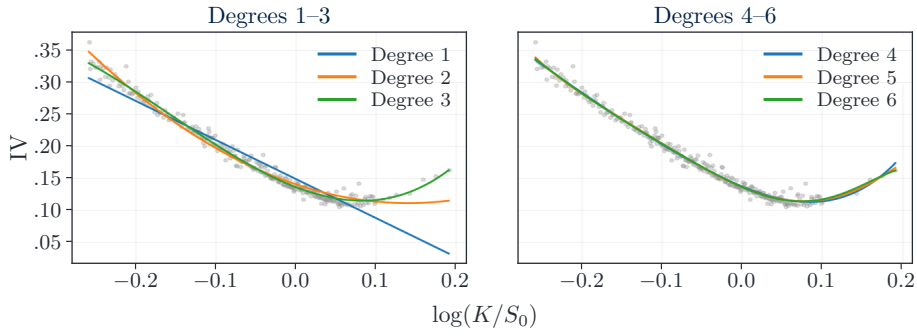
At the 53-day maturity, fit degrees $d = 1, \ldots, 6$ to $n = 288$ observations. Standardise log-moneyness within this slice:

$$z_i = \frac{x_i - \bar{x}}{s_x}, \qquad v_i = \beta_0 + \beta_1 z_i + \cdots + \beta_d z_i^d + \varepsilon_i.$$

Estimate each model by least squares:

$$\widehat{\beta}_d = \underset{\beta_0,\ldots,\beta_d}{\arg\min} \sum_{i=1}^{288} \left(v_i - \sum_{j=0}^{d} \beta_j z_i^j \right)^2 .$$

Q1(b): Fitted IV Smiles at 53 Days



The nonlinear fits capture the right-wing upturn. Additional flexibility gives smaller visible improvements. Training RSS cannot increase with degree, so it cannot select the degree on its own.

Q1(b): Choosing the Degree with AIC and BIC

Let $k_d = d + 1$ count the coefficients, including the intercept.

$$\mathrm{AIC}_d = n \log(\mathrm{RSS}_d / n) + 2k_d,$$

$$\mathrm{BIC}_d = n \log(\mathrm{RSS}_d / n) + k_d \log n.$$

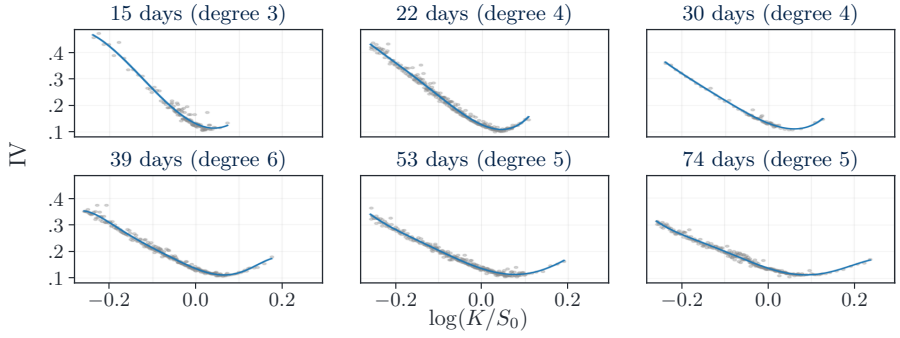
- Smaller values are preferred.
- At $n = 288$, an extra coefficient costs 2 under AIC and $\log(288) \approx 5.663$ under BIC.

Q1(b): Model Selection Results

Degree	AIC	BIC	R^2
1	−1464.629	−1457.303	0.910
2	−1840.736	−1829.747	0.976
3	−2004.621	−1989.969	0.986
4	−2016.111	−1997.796	0.987
5	−2021.959	**−1999.981**	0.987
6	**−2022.813**	−1997.172	0.988

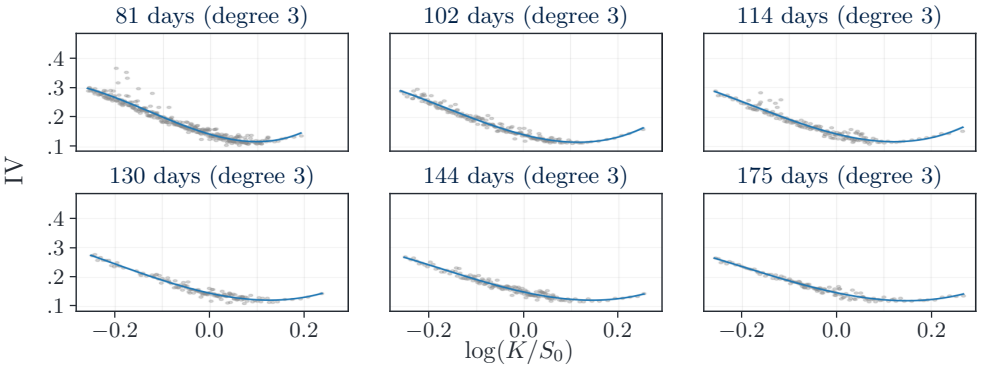
AIC selects degree 6; BIC selects degree 5.

Q1(c): BIC-Selected Smiles, 15–74 Days



Q1(c): BIC-Selected Smiles, 81–175 Days

.



Q1(c): Model Selection Across Maturities

Maturity (days)	N	BIC-selected degree	R^2
15	116	3	0.970
22	274	4	0.988
30	68	4	0.993
39	279	6	0.989
53	288	5	0.987
74	272	5	0.986
81	345	3	0.941
102	200	3	0.972
114	180	3	0.938
130	137	3	0.957
144	157	3	0.956
175	131	3	0.970

A cubic is selected at 7 of 12 maturities. The selected degree is not monotone in maturity; fitted skews generally flatten over common support.

Q2: Ridge, LASSO and Elastic Net

Define the residual sum of squares:

$$\mathrm{RSS}(b, \boldsymbol{\theta}) = \sum_{i=1}^{n} \left(y_i - b - \mathbf{z}_i^\top \boldsymbol{\theta} \right)^2 .$$

Each method minimises the following objective over b and $\boldsymbol{\theta}$:

Ridge: $$\frac{\mathrm{RSS}}{2n} + \alpha \|\boldsymbol{\theta}\|_2^2,$$

LASSO: $$\frac{\mathrm{RSS}}{2n} + \alpha \|\boldsymbol{\theta}\|_1,$$

Elastic Net: $$\frac{\mathrm{RSS}}{2n} + \alpha \left(\eta \|\boldsymbol{\theta}\|_1 + \frac{1-\eta}{2} \|\boldsymbol{\theta}\|_2^2 \right).$$

$\alpha \geq 0$ controls the penalty strength; $\eta \in [0, 1]$ balances the L^1 and L^2 penalties.

Q2(a): Standardising the Predictors

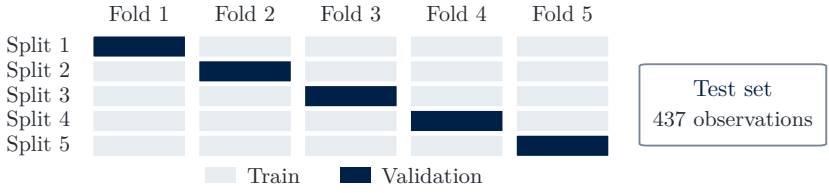
Compute the mean and standard deviation of each predictor using `X_train`:

```
mean_train = X_train.mean()
std_train = X_train.std()
```

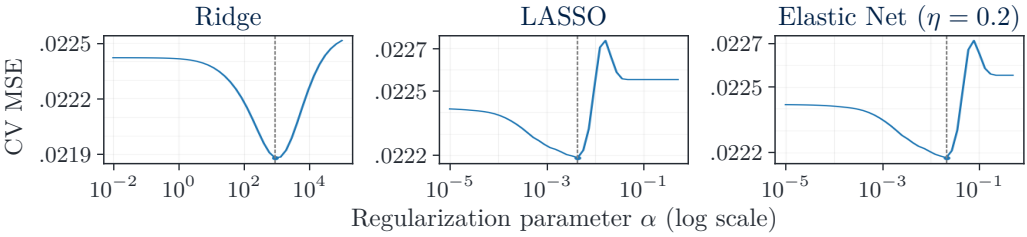
Use the same mean and standard deviation for both samples:

```
X_train_std = (X_train - mean_train) / std_train
X_test_std = (X_test - mean_train) / std_train
```

Q2(b): Five-Fold Cross-Validation



Q2(b): Five-Fold CV Results



Model	Selected α	η	Mean CV MSE
Ridge	893.75712	—	**0.02188**
LASSO	0.00433	—	0.02219
Elastic Net	0.02111	0.2	0.02218

Each score averages the errors from all five validation folds. Compare the table because the panels use different y-axis scales.

Q2(c): Final Test-Set Evaluation

After selecting the parameters by CV, refit the scaler and model on all 1,747 training observations. Evaluate these final fits on the 437 test observations reserved in Q2(a).

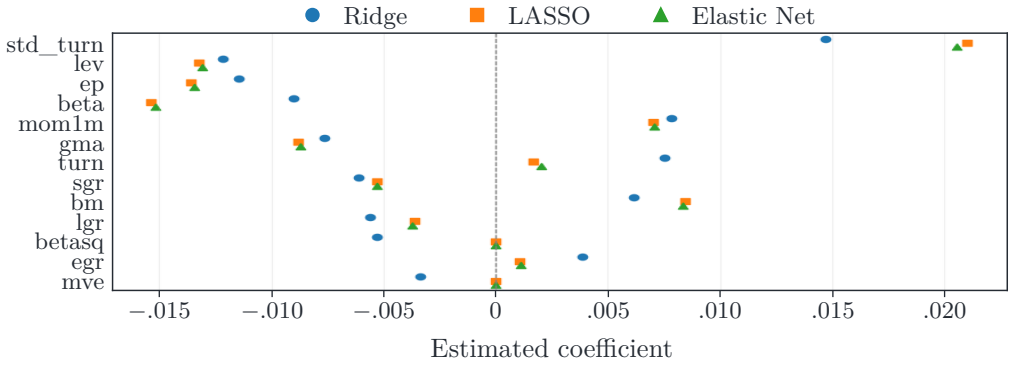
Model	Test MSE	Test R^2	Nonzero coefficients
Ridge	0.0223	0.0205	25
LASSO	0.0222	0.0262	11
Elastic Net	0.0222	0.0263	11

$$R^2_{\text{test}} = 1 - \frac{\sum_{i \in I_{\text{test}}} (y_i - \widehat{y}_i)^2}{\sum_{i \in I_{\text{test}}} (y_i - \overline{y}_{\text{test}})^2}.$$

These test observations do not participate in the five-fold parameter search.

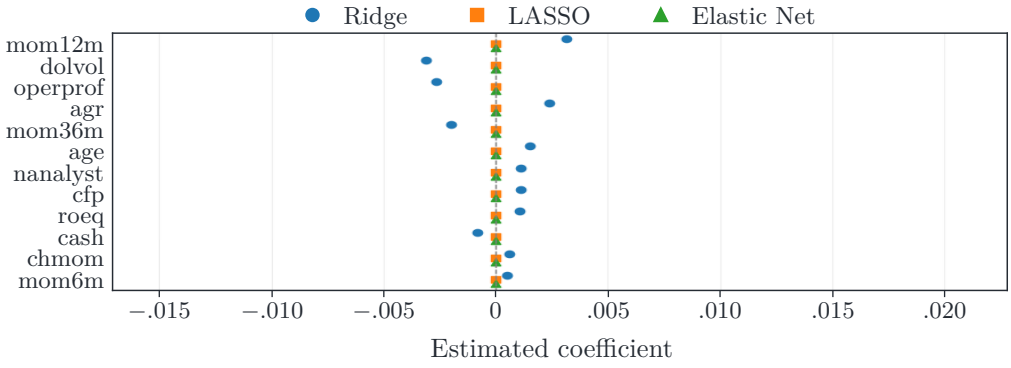
Nonzero means $|\widehat{\theta}_j| > 10^{-8}$, excluding the intercept.

Q2(c): Estimated Coefficients (1/2)



These coefficients come from refitting on all 1,747 training observations with the CV-selected parameters. LASSO and Elastic Net are very close.

Q2(c): Estimated Coefficients (2/2)



Features are ordered by absolute Ridge coefficient; both pages use the same horizontal scale. Retained LASSO and Elastic Net coefficients need not be smaller than Ridge's.

Factors and Firm Characteristics

Factor models: Sheet 1

$$r_{i,t} = \alpha_i + \beta_i^\top \mathbf{f}_t + \varepsilon_{it}.$$

Common factor returns explain stock returns through stock-specific exposures. Examples include market, size and value factors.

Firm characteristics: Sheet 2

$$r_i = \theta_0 + \mathbf{z}_i^\top \boldsymbol{\theta} + u_i.$$

Firm characteristics enter directly as predictors: size, valuation ratios, momentum and profitability.

References: Fama and French (1993, JFE); Gu, Kelly and Xiu (2020, RFS); Dong, Ge, Li and Su (2026, JBES).

Q3(a–b): Rescaling the Predictor

Use the no-intercept criterion $L(\beta) = \sum_i (y_i - \beta x_i)^2 + \lambda\beta^2$, $\lambda > 0$. The first-order condition gives

$$\left(\sum_i x_i^2 + \lambda\right)\widehat{\beta} = \sum_i x_i y_i \quad \Longrightarrow \quad \widehat{\beta}(\lambda) = \frac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}.$$

Scale only x: $x_i^* = cx_i$, with $c > 0$ and y_i unchanged.

$$\widehat{\beta}_x^*(\lambda) = \frac{c\sum_i x_i y_i}{c^2\sum_i x_i^2 + \lambda} = \frac{1}{c}\,\widehat{\beta}\left(\frac{\lambda}{c^2}\right).$$

Preserving the same fitted relationship requires $\widehat{\beta}_x^* = \widehat{\beta}/c$. At fixed λ, the effective penalty changes, so this generally does not hold when $c \neq 1$. Thus, Ridge regression is not invariant to predictor scaling when the same numerical value of λ is used.

Q3(c–d): Rescaling Both Variables

Let $x_i^* = cx_i$ and $y_i^* = cy_i$, with the same λ.

$$\widehat{\beta}^*_{xy}(\lambda) = \frac{c^2 \sum_i x_i y_i}{c^2 \sum_i x_i^2 + \lambda} = \widehat{\beta}\left(\frac{\lambda}{c^2}\right).$$

- OLS: the coefficient is unchanged.
- Ridge: the coefficient generally changes at fixed λ.

Adjust the penalty to preserve the coefficient:

$$\boxed{\lambda^* = c^2\lambda} \quad \Longrightarrow \quad \widehat{\beta}^*_{xy}(\lambda^*) = \widehat{\beta}(\lambda).$$

The squared-error term is multiplied by c^2. Scaling the penalty by the same factor preserves the minimiser.

Q3(e): A Numerical Illustration

For $x = (1, 2, 3)$, $y = (2, 4, 6)$ and $\lambda = 2$,

$$\sum_i x_i^2 = 14, \qquad \sum_i x_i y_i = 28, \qquad \widehat{\beta}(2) = \frac{28}{14 + 2} = 1.75.$$

Data	Penalty	Ridge	OLS
Original	2	1.7500	2
Both $\times 10$	2	1.9971	2
Both $\times 10$	200	1.7500	2

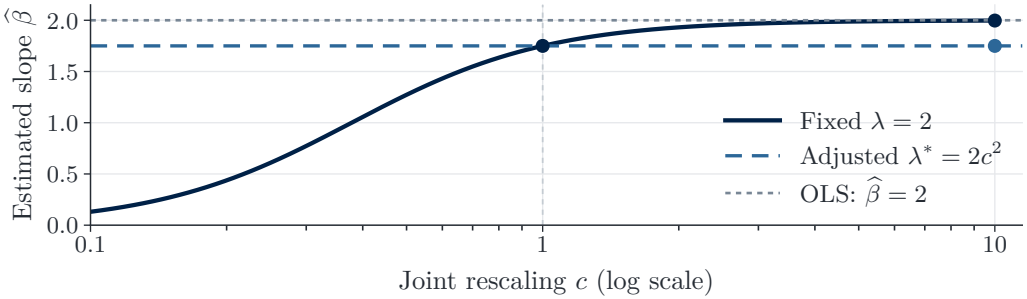
$$\widehat{\beta}^*_{xy}(2) = \frac{2800}{1400 + 2} \approx 1.9971,$$

$$\widehat{\beta}^*_{xy}(200) = \frac{2800}{1400 + 200} = 1.75.$$

Q3(e): Scaling and the Effective Penalty

When both variables are multiplied by c,

$$\widehat{\beta}^{*}_{xy}(2) = \frac{28c^2}{14c^2 + 2} = \frac{28}{14 + 2/c^2}.$$



Setting $\lambda^* = 2c^2$ keeps the coefficient at 1.75 for every $c > 0$.